

ActInf Textbook Group ~ Cohort 5 ~ Session 8 (Chapter 4, Part 1) ~ 11/7/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 8 (Chapter 4, Part 1) ~ 11/7/2023 | 53:32

all

right hello

everyone coort 5 it's our first

discussion on chapter 4

so does anyone want to give just

any thought on four

overall just any any aspect of what they

expected or got from it or however far

they got or some section that was

just interesting to

them I have a

question

um

so one kind of generative model that I

am familiar with is um one way of

representing a gener model is using

markup decision process or partially

observable Markov decision

process uh this is for

discrete

um the discrete case of generative

modeling I'm

wondering what else what other what are

other very popular or

common uh structures that are used to uh

write down or Express generative

models anyone want to give a

thought

I would say one alternative way to write

down parts of generative models is the

continuous time but are you asking is

there another way to represent discrete

time generative

models other than with a

PDP right to begin with because of

course there's the the whole
other uh area of continuous time
but um there's one important aspects to
the
whole um question
of
um of basian inference which
is or some some agent some organism
that's implementing basian inference is
how does it uh not only learn the
parameters of a generative model but how
does it also learn the structure of that
model um and
this is connected for example to
questions about nature versus nurture
and and uh development in in humans or
animals like what are the um the initial
conditions what kind of um either
inductive priors or like um structure
biases doeses our neural
architecture uh provide us with from the
very beginning
and whatever it doesn't provide us with
how how do
we um figure out what kinds of structure
is appropriate for for
modeling a particular
domain all you have any
thoughts
uh well first of all uh about the
initial uh states of any system uh that
I mean it it doesn't always um in active
inference modeling we don't we don't
always assume U any agent like tabul
laasa like I mean we don't always begin
from scratch so uh in most cases we
necessarily impose some initial
conditions uh in order to make it more

efficient to model the exact behavior of
or phenomenon of interest uh because
otherwise we won't be able to
um to model the whole process I mean all
the emergent properties that um the
agent have acquired the the agents have
acquired through their lifespan uh so it
is uh it wouldn't be feasible to try to
um put everything in in such an
encapsulated form of um generative model
in order to Encompass uh the agents uh
entire u i mean experience and lifespan
so it is necessarily uh more focused
than that uh and uh IT addresses a
specific problem uh we're trying to uh
we're trying to model so um I guess uh
it's
not uh a priori it's it's not
something a priori uh and it isn't U any
prerequisite to um model uh any
situation but uh when it comes to
how an agent acquires those uh initial
learnings those initial States well
obviously there are many many ways to uh
address that uh problem U with with
various perspectives but
um of course F or active inference uh in
my opinion is not um a theory or a
framework that can answer every um every
question regarding uh to the behavior of
the agent uh in every possible situation
so we need to look for uh other or other
granular problems um in order to uh
account for the lifespan experience of
an
agent got it uh thank you
so um just to kind of reiterate and
clarify this your response for myself um

in my mind I kind of related a bit
to on the one hand this distinction we
we often need to make and keep in mind
between an uh a realist interpretation
of active inference and an
instrumentalist one uh where the latter
is us taking this this mindset of like
okay uh there there are agents out there
in the world there are things out there
in the world and and they do stuff uh we
can't be completely sure of exactly what
they're
implementing uh mechanistically but but
we can develop a framework for uh making
some SE sense of it for ourselves and
part of the way that we make some
progress on that uh making sense project
is to use our own um suspicions use our
own uh Paran knowledge to to write down
or Express or postulate some some
structure to postulates some generative
model uh and then we can we can active
reference as a template where we can
input that generative model uh perform
inference on it and then uh perhaps
eventually make some
predictions um about the behavior of a
system and compare that to to what the
system is actually doing but but what
you're uh something that you're pointing
out is that within the context of the
framework uh there isn't necessarily
some
uh uh procedure
for figuring out okay if you didn't have
this uh these priors about what what it
is that the generative model could look
like uh and

and I suspect ECT we suspect that uh for example humans don't come with all of the the structure that they need uh embedded from the moment they're born in order to

model uh the world maybe they have some structure built in but they they acquire more of it and learn more of it over time um but active reference is is not doesn't have all the tools um to to perform that learning of the structure you that's also an area of of work

that's you can do but it's not

explicitly part of active

ref uh just to clear up a little bit um

here because you see even in the the

very fabric of the first principles of

active inference namely the free energy

principle so the free energy principle

uh assumes the existence of quote

unquote things and then try to

characterize what would the existence of

those things constitute okay so it it

doesn't necessarily uh provide an

account for uh I mean uh how the

existence occurred how how I mean how

how those emergent uh properties arise

from um from from lower causalities

of course it can describe uh those

causal Relationships by assuming that

those uh uh emergent phenomena uh have

happened uh already so uh it takes as

very first as the very first step and as

an initial uh as an initial U State the

very existence of those things right uh

so when it comes to uh Pro U modeling um

an agent's Behavior within active inference framework I believe the same mindset applies uh here as well so uh it doesn't necessarily account for the uh how questions but rather to um it it tries to characterize the what questions uh I'm not sure uh if it clears up the I mean your question here or not but um it's it's not an account for um it's not an account that uh tries to uh let me put it this way um if if we have this system at our disposal uh that we we try to understand um the its mechanism or or um the way it behaves active inference provides uh a comprehensive tool to uh write down uh the behavior of the agent uh within specific uh mathematical technology but it doesn't necessarily account for the um emergence of that agent it doesn't necessarily account for how that agent uh emerged at the first place uh so that's uh one of the misconceptions about free energy principle and uh active inference because some people uh think that uh free energy principle being the theory of every quote unquote thing would necessarily entail uh accounting for the emergence of uh those ontological entities uh but uh it it's it was never uh the claim of free energy principle so um yeah it takes as uh it takes the existence of those things um as the given and then try to describe those uh Behavior thank you um inks then

Andrew hi I have a separate question but now I got involved in this so but I do want to understand so if we were to model like super deep thick temporal modeling of a person as they're born and they go through their life if someone was sitting there active inferencing everything that they're doing that would account for the development of the faster prior updating of priors and the slower updating of generative models right that would somehow build like a human with their understanding of the world not build but model how they become themselves okay well I mean to to the coarsest approximation yeah you could make a generative model that included decades would that mean it would give useful insights or interface with empirical data not not at that course graining could it be developed to be richer and more useful and meaningful interesting question but I don't know does anyone else want to give a thought on that well I would find it useful I do have another question though I was just okay yeah okay sorry so this is um from the book page 77 um where the figure 44 describes the basian me basian message passing so my question there is the the black circle one the one that says intuitively current beliefs about States under each

policy at each time are compared with those that would be predicted given beliefs about States at other times so this is the the statistician inferring by comparing different predicted States at different times does that mean is basically time flowing left to right and you're trying all these different policy policies or are these alternative policies about something you can do right now and you can do different things about it and you're trying them out in your mind is this time flowing left to right or are these in my mind like if I do this this will happen if I do this this will happen and I compare them in my mind almost parall like they are counterfactuals to each other thank

you yes

Oli uh okay so there's a distinction and active inference literature between uh the time as uh t as the letter T and to so uh more often than not when active inference researchers use the letter t for time uh they uh they mean the flow of time as we know it I mean the intuitive uh time or our perceived time but uh what what but when they use to it doesn't necessarily refer to the flow of time but rather uh to uh to the steps uh that an agent uh would take in order to achieve a goal so uh to in a spec in a particular moment is not necessarily uh uh I mean the time interval between two tows are

not necessarily equal right so uh it it
tries to uh model the specific steps of
uh the decision making or the action
taken uh to achieve a particular
goal thank you that's one aspect I think
there's does anyone else want to give
another thought on this and I clarify my
question is are there counterfactual
policies that only one will get
implemented or is it step after step of
of enacting like a project of long a
strategy of many
policies
so in the pseudo code usually as
implemented during the
policy updating so going from the policy
prior to the policy posterior that's
using the expected free energy so that's
when different policies are being
considered so policy however long it is
it's going to start with your next
action and then include that many steps
out so then if there's you know four
options at each step four affordances so
you're going to do policies of step two
so there's 16 so then in the pseudo code
it would have there's 16 policies and
then there's the prior so if they're all
equally likely then they be all even
then all of them are iterated over in a
loop and then while each one's getting
iterated over it is rolled
out and then then they're all compared
at the
end resulting in the
updated
distribution and then so let's just say
it's it's um 80% for the first option

and then 10 and then five then then a few so then you could draw from that distribution and choose to do that or you could pick just like the best option which is actually something that um Magnus discussed in this uh live stream

55.1 which is about this message passing because like there there it's kind of a lot to it but also it's a very important relationship and also one that's not really addressed in the textbook just a few times kind of dropped

in uh okay Andrew and then anyone else yeah I just wanted to give my stab at uh the prior question um it seemed to be asking what active or the free energy principle is or isn't and um me it it's a set of constraints it's it's telling you what open functions and Funk do namely your C's your Q's if I am correct um can and cannot do if they are to be an entity that persists spatial temporarily um so it it doesn't tell you what those functions are they it leaves them as three variables but it constrains their behavior

so so it I guess it's a different way of thinking about things than um thinking about like what is force equals

mass times

acceleration it's not showing transfer of energy it's saying where energy is constrained to go if something is to persist over time

yeah that's very interesting um also

this this is one one paper the title
kind of summarizes this very simple yet
perplexing argument by axal
constants it's not about what it takes
it's about what took you
there
so it's pretty interesting um
ra yeah can you elaborate what's uh your
interpretation of that title
is of what is of this title it's not
about what it takes it's about what took
you
there I think here is said really well
um so just to to generalize it um if
something exists right now under the
right conditions it's because it's done
something like minimizing its free
energy so it's as if it's
used variational free energy however the
question of whether it will exist
tomorrow cannot be settled purely by
resorting to the fact that it will
minimize its free energy to get
there so something could
just all just you know it's I I don't it
may also be like saying um like past
results is not future performance or
something like things could change in
niche so it's
it including it minimizing free energy
right to the
end so there's kind of a few other
aspects but but
um yeah and and related to as Oli
mentioned earlier like conditioning on a
thing
existing by by choosing that kind of
boundary of framework then there's

questions that aren't being addressed
and ones that are but yeah right so just
to clarify my original
um so definitely we in order to be able
to apply any of this uh of these ideas
we take for granted we take as uh as
assumed the existence of mar conf BL kit
which is what gives our definition of a
thing um my question is about uh so
already assuming that a thing exists and
it has its its
boundary uh
internally it is
representing is encoding some uh model
of the generative process that's going
on
outside uh in the world and it um uses
perception and action
to um make sure that that model is as
consistent as possible with its
observations
um in the world so my question is about
what's inside that generative model um
for example in the case of a pomdp which
is one way of uh
representing uh a discrete time
generative model uh there's some
structure to what that PP looks like the
way that we write it down and what what
are the parameters um and I need to
become more familiar with with the
actual degrees of freedom so to speak
the different things that you can uh add
or or take away to construct a different
varieties of of of a model but but let's
just take that
PDP um
category as an example and the way that

I am thinking about it right now what
what you have is as
you're um incorporating this this
generative model
you have some parameters and uh over
time you update the parameters of
through perception you update the
parameters of that generative model to
be more consistent uh to to reduce uh
your free energy to minimize your free
energy but my question is about the the
the structure of that generative model
not just the the parameters but for
example if uh if it's a a basan network
um that is describing the causal links
between some uh hidden variables some
causes leading causes in the world and
like the observations that you're
receiving that basan network has some
structure there are different nodes and
connections between that and uh somehow
you have to figure out what should that
uh Network look like uh should the
connections between nodes be and what
should each node
represent um and and so I'm
wondering if within the context of
active inference there
is uh there's a procedure there is
procedure for
exploring uh that structure or for given
a thing that already exists how does it
the its generative model evolve not only
in terms of its
parameters uh but more
um uh generally in terms of its
structure maybe this is related to the
idea of hierarch models which uh I

haven't uh learned much about yet
but yeah maybe I hope that to Claris
applies a bit more my question yeah two
short answers first there's structure
learning that is the whole question of
modeling
and that that's what the question of
modeling is
that's why it's just not done just by
thinking about it because you have to
fit the base graph to be capturing
whatever factors you're
modeling but also this paper and some
other ones in that area approach the
structure
learning and also um in live stream
54 with the learning structure versus
learning discussion that that um Toby
uses excellent thank you did you link
these somewhere um uh yeah just in in
where the notes are the notes for
chapter four four thank you so much yeah
all right any other like thoughts on
four or some section that some one found
interesting
if uh no one else has a question about
four I have a question
about uh inference algorithms but but I
want to give everyone the chance
to to ask your
questions
does anyone have like any overall
thought on for yeah okay
Andrew sure I I was still kind of
reflecting on the the question earlier
about message passing I feel like I'm
going to mix some things up here but so
that's back on figure 4.4 and I was just

thinking about the question of
counterfactuals
um because I mean if um if an agent were
to and you'd have to of course like
model them you'd have to construct them
in such a way that they were able to
engage in planning which I suppose would
involve minimizing expected free energy
G um so in that in that case whenever
one is engaging
in you know planning if we could model
that um then I guess is someone I mean
they'd be comparing counterfactuals in
that case right they'd be comparing
different potential policies that they
have not already engaged in or or
potentially are already underway um in
some way but like you know maybe you're
so far along the way and your series
sequence of actions associated with a
particular policy or you haven't started
it yet and then from there the idea is
you would go with the policy that has
highest probability is it right you You'
basically you'd go with the policy that
minimizes expected free energy which
would also like whenever we look at that
in terms of a distribution over policies
it' be the policy that has the highest
probability which you end up necessarily
going for um I'm just trying to because
I'm slowly learning how important it is
to specify when you're modeling like all
these factors and things can get a
little blurry whenever we start speaking
a little bit more philosophically or
from real life experience I'm just I
don't know if all of that made sense

right but U just throwing that out

[Music]

there just a quick uh

clarification

question uh you brought up

contrafactual in relation to action

planning and then uh Daniel wrote down

here planning considering future times

steps is all

counterfactual uh what that brings to my
mind is that the way

that um we're thinking about inference

sorry planning in active inference is uh

Imagining the

outcomes of different

policies uh and rolling those out uh

into the future and then Computing the

expected free energy over those roll

outs for each

policy um and then you kind of score the
different uh

policies uh according to their expected

free energy and this kind of gives you

uh a distribution over

which uh policy you're you expect to

take or which policy you believe you're

you're going to take um but but it seems

like you you maybe like you sample from

that distribution so it's not like you

you just uh take

um the policy with with the the the

lowest expected free energy if that's

not true please

someone dispel that yeah right yeah all

planning is that's one kind of

counterfactual counterfactual just being

um considering something something that

is not a factual like observable so

anything about the future time steps is counterfactual um or like alternative pasts but then considering the future it's all counterfactuals and then this is also kind of from figure

5.4 um this is that section of um taking in habit and then either passing habit through the PO prior or updating it according to expected free energy and then but whether whether habit has passed through or there's an update um then the most likely path results or that distribution is sampled

from yeah Jesse so these are two yeah these are two possible selection uh methods either maximum um posterior or or sampling those are two methods of policy selection given a policy

posterior I guess for me uh what I'm attempting to do is I mean because quite frankly it's all a bit over my head but um trying to pull this back from like a lot of this conversation

you know we're we're trying to apply this to conscious processes which I think makes it a little bit more difficult in some respects because there's so much you

know philosophical baggage um and so in the last section of this chapter even though I don't really understand the application of like the um no I'm sorry it wasn't the Lan uh approxim yeah the leace

approximation uh on same motor reflexes and trying to think about this like even

thinking about the message
framework uh in a purely physiological
Paradigm where the message passing is
say you know the
sensory response from the you know the
doctor hits your you know under your
knee with a little rubber hammer and
either you're going to have a reflex
response or you're not right and so in
that in that respect if I try to look at
the the counterfactuals or the policies
not as
some you know
conscious
um process but as a purely phys excuse
me my throat but is a purely a
physiological
one it just kind of I think for me makes
it a little bit and maybe I'm on the
wrong path and just Mis misunderstanding
it but for me it's just making it a
little bit
uh more simplistic I guess to to deal
with uh all the all the symbolisms and
the idea of gradients and the idea of
it's not like we're consciously always
doing this and and you know undergoing
the statistical process but this is you
know essentially a way of understanding
the probability that you know given
certain factors and
parameters you know your knee gets hit
with a little rubber hammer and you or
you don't so I guess I just want to kind
of throw that out there
for possible
Direction yeah I mean it it's very true
and relevant that that not not

everything Maps or just people may have wildly different maps of what features or psychological experiences they map to like different intuitions about these Concepts so it's a huge space um Andrew then raw um I I don't want to be redundant uh Daniel pretty much summed up my point but it's yeah I think it's more about just thinking about it as far as like you're a like take the perspective of someone who's modeling um think of like you could produce a model of like a person or or some other phenomena whatever agent you're looking at and you could include planning as like an entire sort of Dimension so to speak or or layer to your model but that could just entirely be for one I imagine most people who are trying to model you know uh reflex arcs and looking at yeah the movement of joints in response to you know the pressure at this particular spot probably from an active inference approach I imagine would not include planning in the modeling process meanwhile of course if you're trying to model someone thinking and you know Consciousness or however close you want to come to that I think last week we discussed the the neurop phenomenology paper I could drop a link to that again but like that kind of modeling certainly you could you could include planning as um as a dimension and and I bring this up again because it's just that's what I'm slowly learning with active inference is like while I enjoy the

philosophical discussion I think it's
super important to just think about like
from a modeler's perspective even though
that kind of like can remove some of the
magic of of the process but it's um it's
you choose what you're going to include
or not include in the model I suppose
right um I think I I just hope that that
clarifies anything it was clarifying to
me to start to understand
that yeah just to connect to um Thomas
pars guess
stream um and like a little bit of the
development of different generative
models because chapter 4 is where the
generative models being described seven
and eight will go into more detail into
the discreet and the continuous time but
chapter four is kind of like just
considering generative models um
generally so the early implementations
were in continuous
time then people were thinking about how
to get sequential
Dynamics that's discussed in this
textbook with these Predator pre type
models so that is like where there was
repetitive curse of
writing so it was
doing like repetitive
structures but it was in a continuous
State space of time um then they started
going more into the explicit
planning explicit like the pdps kind of
tying back to the earlier discussions
that we had so
um and
then later um The Continuous models were

used at the more like sensory motor
layer and then the discrete state space
models at higher levels of
decisionmaking

just want to make that point Thomas
explains more but
yeah any other like thoughts on
four yeah I'd like to um chime in just a
bit

so I think it's worth noting
that you this whole active inference is
um is still very much in its
infancy and you there's a lot to work
out I mean you know when we talk
about

um and you to Jesse's point is that you
there there is no Authority here I mean
we're talking about how humans model
their

lives and so

um the assumptions that not only the
modeler has but also the person has is
going to um you know bear on the
counterfactuals and so you know to
produce a model of a person you know we
can't

really discount the fact that what
they're perceiving is not
reality

so I hear a lot of it assumptions that
so I I question these assumptions and
maybe it's because I'm coming out of
worldly mapping but you there's just a
lot of assumptions that you bring up
questions for

me um as to the accuracy which is yeah
the intent of the whole project is
to become more

accurate thank you

Susan one approach there because I I I

really agree with with a lot of what you
raised um by using similar terms and
just being clear about what terms we're
using then some Expressions can be
evaluated and then and just sort of like
very carefully

understood and then different people may
still make different claims about the
same topics C certainly that's what
discourse

is right that's a very

interesting aspect of this poort and The
Institute more broadly which is as I
mentioned Susan this is

very um stof thee art like a young young
kind of
project um

and The Institute is part of the group
of people who like everyone here in
trying to understand the framework is
participating in developing it and
carrying it Forward uh but also
something that's I know that the
Institute is really um working on uh and
keeping in mind is how to to
optimize uh this process of of doing the
science of thinking about the ideas
uh collectively and as Daniel just
mentioned part of that is uh to the
extent that we uh engage with each other
more often and have these discussions
together uh and use um the most precise
language that we can the
ontology uh and become more familiar
with the ontology then we start to
become kind of like uh almost to

approximation uh I have like a
mathematical uh approach to thinking
about it because in math you have very
clear definitions about uh for
everything that you're working with so
so this is very helpful that um that
we're we're becoming more familiar with
the concepts and the terms and then
using them um in a shared way in a way
that hopefully makes sense to as many P
people as possible in every
session but it's also really excellent
to have people who um are maybe less
familiar with some of the technical
details because uh then you can stop uh
others and say like hey that's um isn't
completely clear to me or I'm thinking
of it from this more intuitive
perspective can you um explain how my
intuition laps into
your to the terms that you're
using yeah thank you um Andrew
yeah quick follow on to that point um I
was hoping the ontology on our um uh
what do you call it our epistemological
space had a lookup table for the um
equation
symbology and I didn't find that but
it'd be really helpful if there was like
a word
or something I can do for lookup
card to um help embed that antology uh
built into the active inference
textbooks uh equation into my
brain H it's a great idea it's
definitely something we've considered um
there's several steps that we know we'd
have to do so like in our literature

analysis from the end of
2022 we annotated the number of
equations in each paper for for several
hundred papers obviously we have that
information for textbooks so determine
where the equations are then just like
we have for um some of the equations but
not all of them um develop the natural
language descriptions using the ontology
terms that natural language description
would be including all of the terms that
you need to include make the mapping
between the symbols and the ontology
terms I mean it's a project that that
ontology will include and other features
pending

slash people who want to do
it thanks for pointing that out yeah but
it would be epic and and if you want a
sub page or if you want a form to do it
in let's figure out how to do

it

yeah

Susan

um so if I'm premature in asking this
question just you know just say so and
we move on but before equations would
become accurate wouldn't wouldn't it
be um a matter of um pattern languages
would be more of a broad scope to be
able

to um you

know audit the equations

Spa that's a very interesting

yeah lot lot of ways you could say it

the equations are being evaluated

according to kind of standard math so if

we consider that as a pattern language

like math is a pattern symbol
set then and also the equations
equations are
usually trivially
true because it's they're codifications
of assumptions basically yeah they're
manipulations of of what is brought in
to the
formalism right so then that has a
certain epistemic status it's not
nothing there are theoretic iCal and
mathematical findings that that are they
are real
findings but that that doesn't justify
or or imply direct application of to
anything of of
anything okay thank
you yeah the on like making it more like
a living textbook with rendering
um equations with the natural language
or change making the variables
compatible or using the computer
variables
um those could help a lot otherwise
there's a lot of variability in which
notation are used and to what extent
notation is
described like hopefully to a limited
extent in a
textbook but still there's a lot of
notation in the textbook and it it kind
of like shifts among the CH the chapters
just bringing in more variables not
necessarily like using one core set of
variables which would be hard to do but
that's not what they
do so one more
question since we're at the top of

the um would it be premature to um
announce that we're we have a special
guest that might challenge our
assumptions on um
physics

Daniel I

mean we'll we'll announce it when it's
organized but yeah that sounds
good anyone like have any thoughts on
chapter 4 or like or what they want to
ask about next

week

anyone have any other thoughts on four
inks I just want to say I love this
equ they're not equations but the ones
that look like waterfalls that go in
different patterns it's very inspiring
to see them and just to Express
gratitude for you guys you are Saints to
do this with us to just sit here Daniel
and try to figure out what the hell I
mean when I say that and want to help us
understand like I don't know that you
guys get enough appreciation and
gratitude so just saying
thank

you yeah it's awesome that people come
out and and care and it's interesting
topics and even without even
understanding only at this level of
zoomed

out it's still is pretty

cool I want to second that thank you

Daniel yeah thank

you 100 % it's amazing all you

guys yeah everyone updating generative
models learning about

it we can only do like the tiniest

amount in the synchronous times but you
know

yeah okay cool see you all next time for
part two on chapter four

thank you everyone great to see bye bye
bye